GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 08



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Recap of Previous Lecture



Topic Mixtures Alligations

Topics to be Covered



Topic-1 Number System

Basics of Number System

405

Whole Number : 0, 1, 2, 3 - ∞

Natural Number

∞

Even

$\frac{12}{6} \Rightarrow 2$

6) 12 (2
12
--

$\frac{15}{3} \Rightarrow 5$

3) 15 (5
15
--

$\frac{0}{9} \Rightarrow 0$

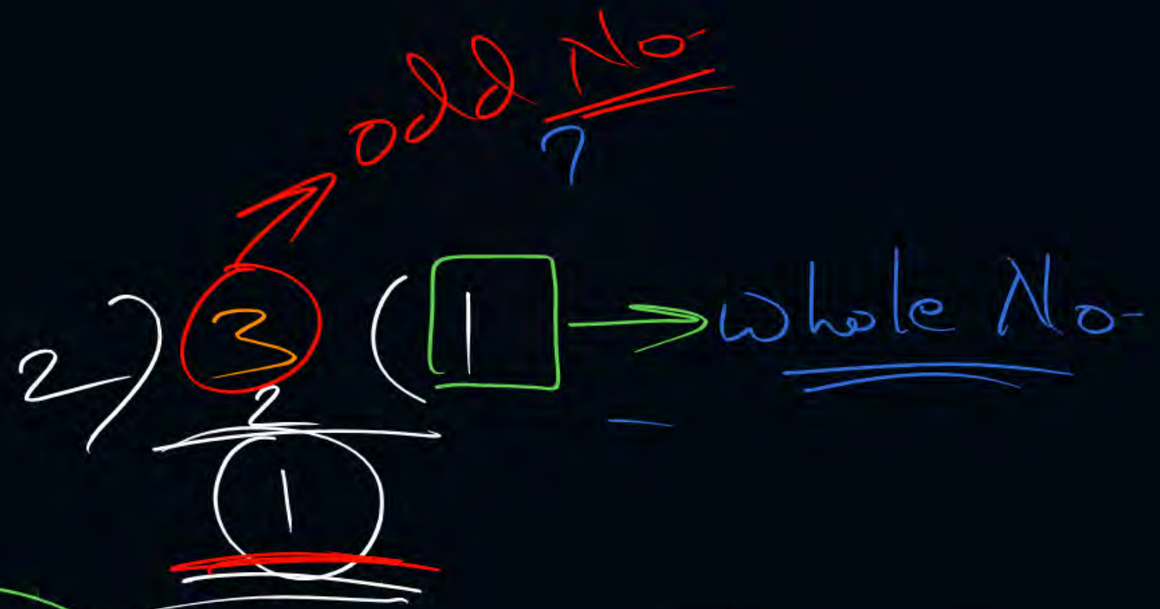
9) 0 (0
0

$\frac{13}{0} \Rightarrow \infty$

0) 13 (

Even

Odd



Prime

1 0 ONLY

1 + 1

PRIME

Only two factor

NON-PRIME

More than two factors

Factors

6 => 1, 2, 3, 6

Multiples

?

6 => 6, 12, 18, 24, 30, 36, 42
48, 54, 60, 66, 72, 78, 84, 90
96, 102, 108 —— ∞

RATIONAL

$\frac{p}{q}$ ✓

$q \neq 0$

$9\overline{)382}(42.444$
36
22
18
40
36
40
36
40

$3 \Rightarrow \frac{3}{1}$ ✓

$3.8 \Rightarrow \frac{38}{10}$ ✓

$0 \Rightarrow \frac{0}{1}$ ✓

$-8 \Rightarrow \frac{-8}{1}$ ✓

$42.\overline{4} \Rightarrow \frac{382}{9}$ ✓



$x \rightarrow 42.\bar{4}$

$3 \Rightarrow 3.\bar{0}$

$10x = 424.4444\ldots$

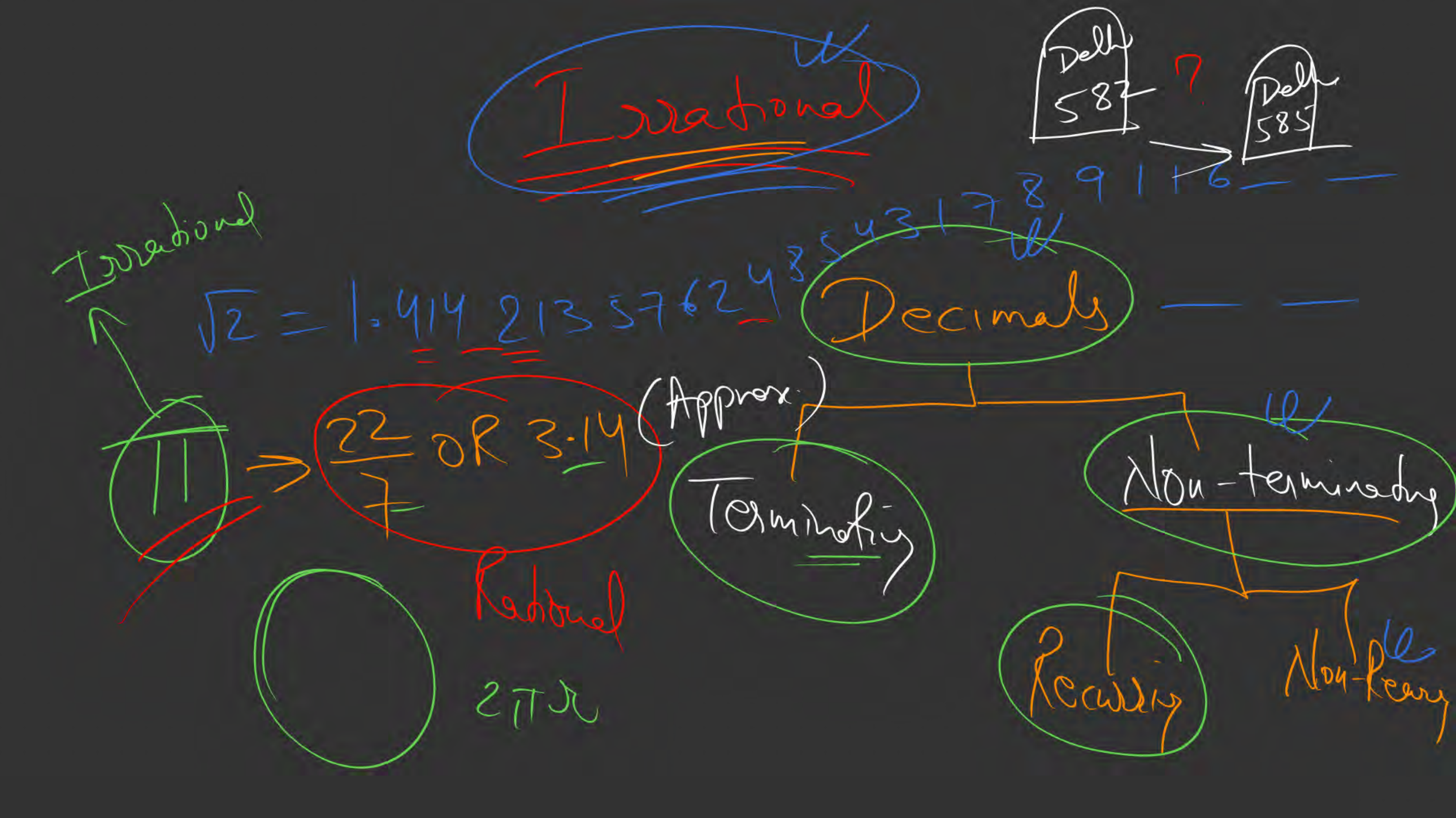
$-\ x = 42.4444\ldots$

$9x = 382$

$\frac{45}{9} \rightarrow 5$

$\therefore x = \frac{382}{9}$




Irrational
Delhi 587
?
Delhi 585
8 9 1 6
Irrational
√2 = 1.414 213 562 4 3 5 4 3 1 7 8
Decimals
π
22/7 OR 3.14 (Approx)
Terminating
Non-terminating
Rational
Recurring
Non-Recurring
2πr

$7\overline{)22}\,(3.\overline{142857}$

21

10

7

30

28

20

14

6

60

56

40

35

50

49

10

[MCQ]

#Q. Find the answer of given expression: $1 - 2 + 3 - 4 + 5 \ldots\ldots\ldots 2025. =$

$-1 \quad -1 \quad -1 \quad 2024$

$1! \Rightarrow 1$

$2! \Rightarrow 2$

$3! \Rightarrow 6$

$+2025$

$= 1012$

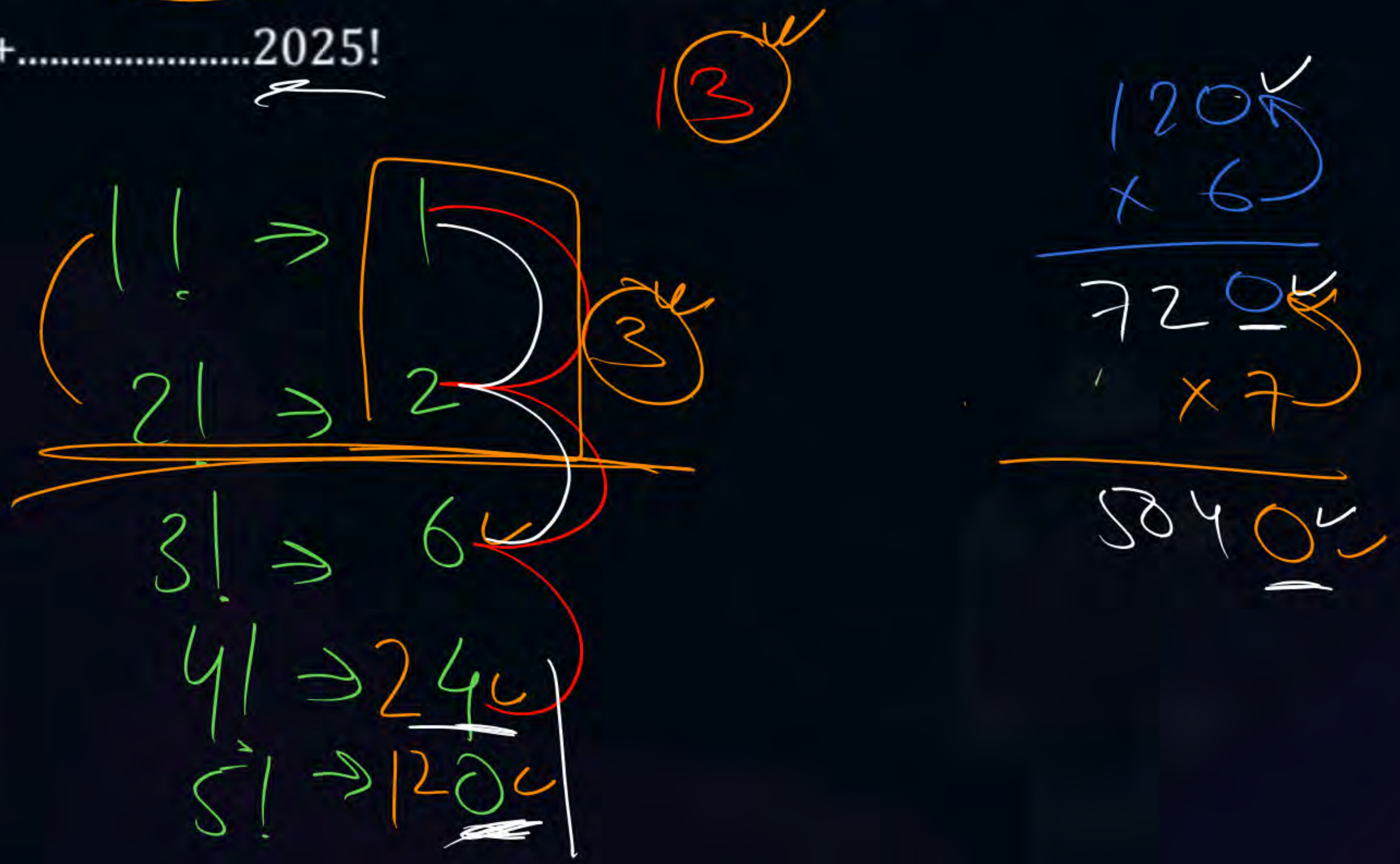
1013

1013

[MCQ]

#Q. What would the unit digit in the answer of given expression:

$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2025!$



[MCQ]

#Q. What would the remainder when the answer of given expression is divided by 3?

$$1! + 2! + 3! + \ldots\ldots\ldots\ldots 2025! = \text{Ans}$$

Ans → Divisible by 3

(A) 0

(B)

(C) 2

(D) Can't be determined

3) Ans ( )

R = 0

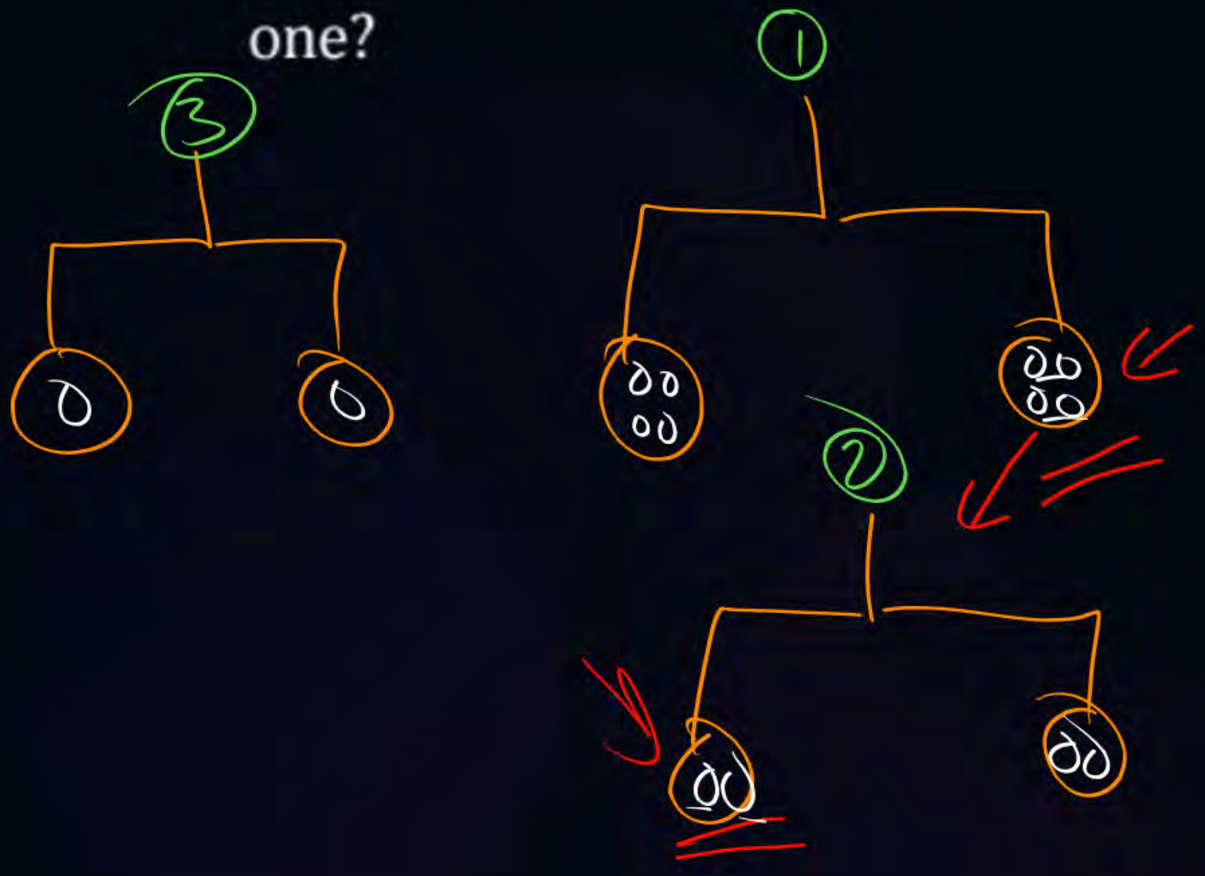
Bag → Flour

↓

Rice bag → GATE

[MCQ]

#Q. There are 8 balls of which one is defective. Given that the defective ball is of less weight and remaining are of equal weights. What are the minimum number of chances a common balance is to be used to find the defective one?





[MCQ]

#Q. A bag consist of 48 red colour balls, 16 green colour balls, 12 yellow colour balls, 11 black colour balls and 11 white colour balls. How many minimum number of balls are to be taken out from the bag randomly so that we get at least two balls of same colour?

6 balls

[MCQ]

#Q. A shelf consist of 40 socks. 40% of these are black and remaining are white. How many minimum number of socks are to taken out from the shelf randomly (blindly), so that we get atleast two white socks?

40 ✓

2 white ⇒ 18 ✓

Black: 16

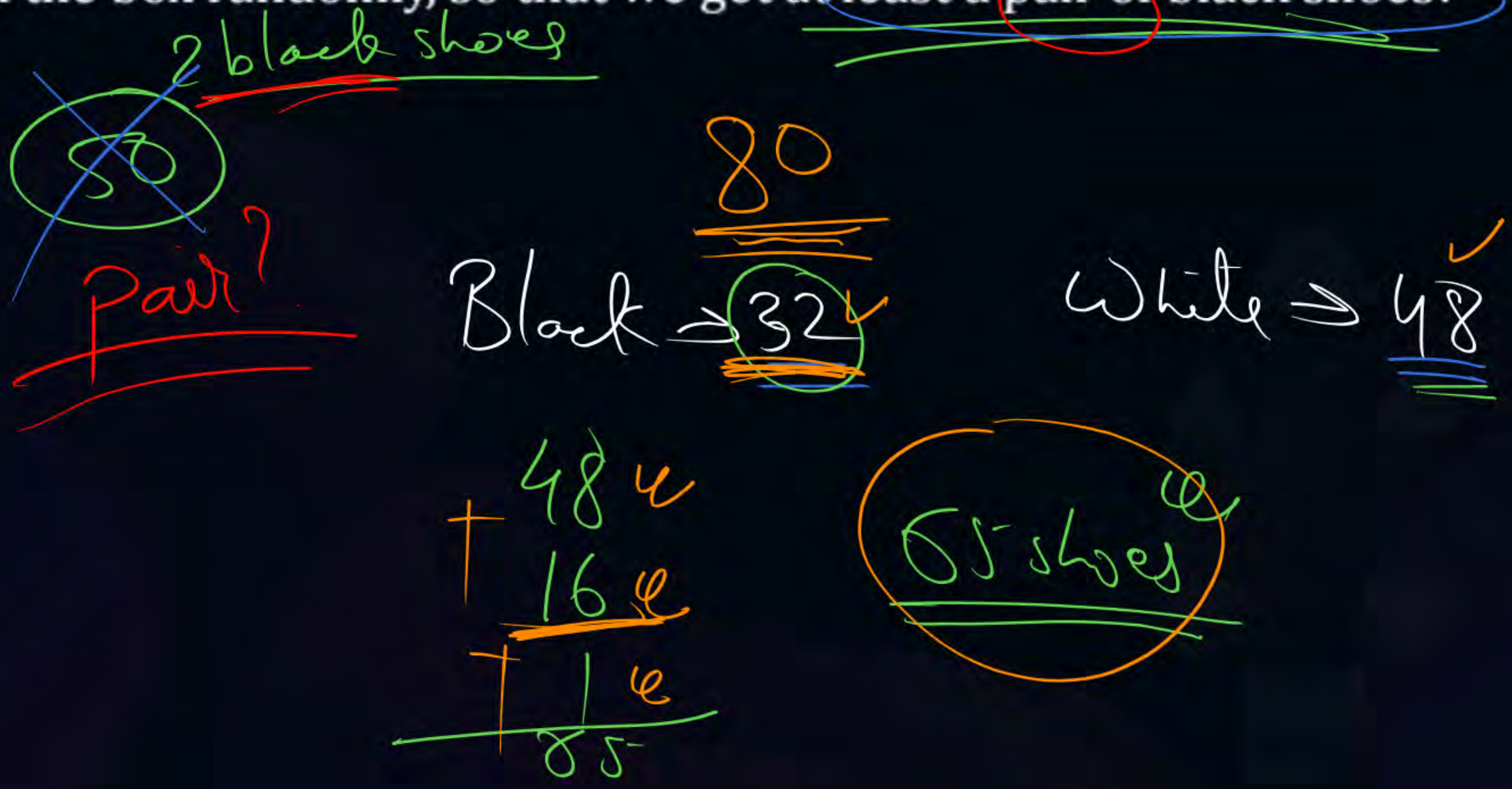
White: 24

Atleast 2 black

26

[MCQ]

#Q. A box consist of 40 pairs of shoes of equal size. 40% of these are black and remaining are white. How many minimum number of shoes are to taken out from the box randomly, so that we get at least a pair of black shoes?



[MCQ]

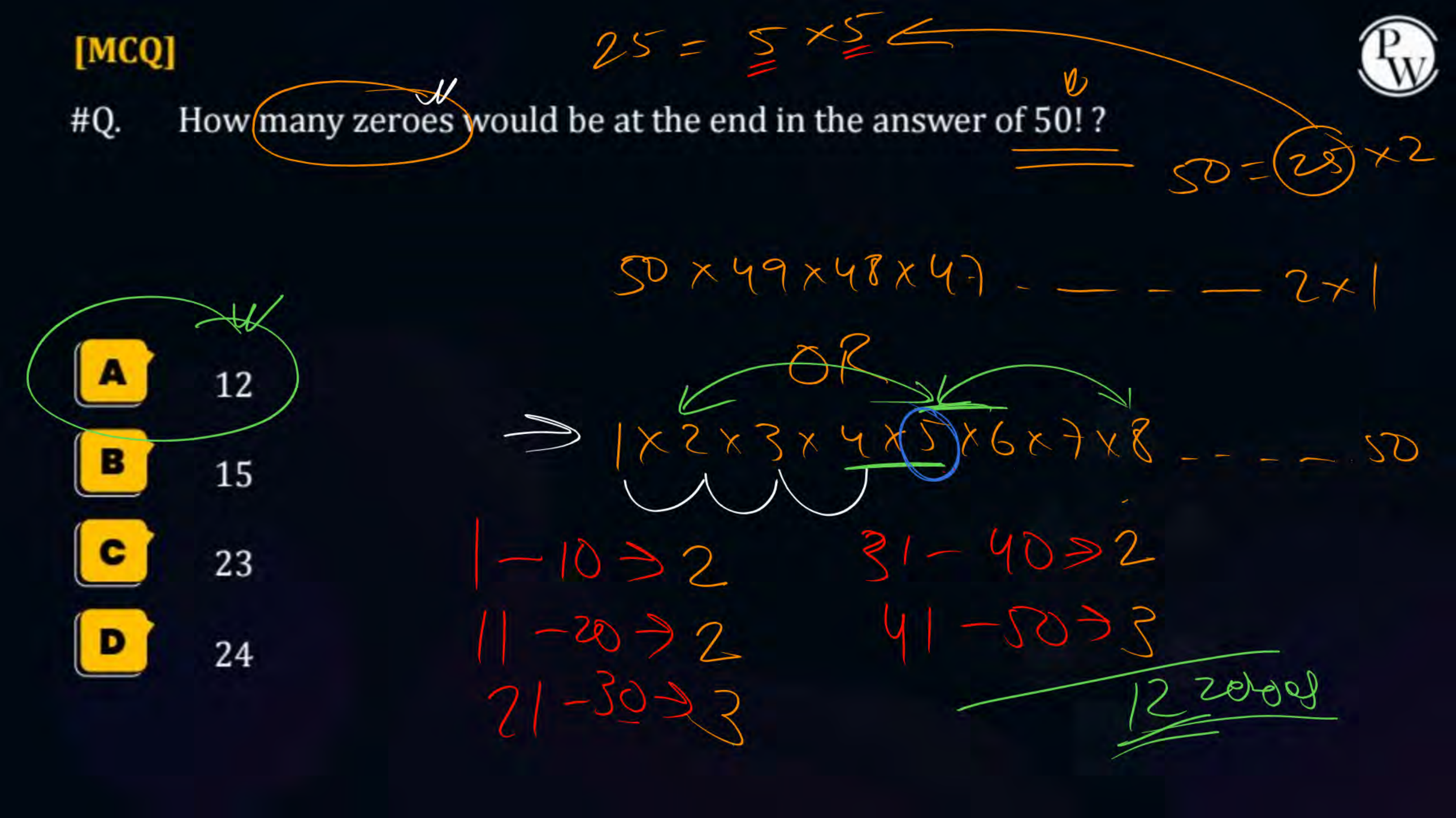
#Q. How many zeroes would be at the end in the answer of 50! ?

A. 12

B. 15

C. 23

D. 24



(prime) Short cut ✌

$\frac{50}{5} \Rightarrow 10$

$+$

$\frac{10}{5} \Rightarrow 2$

$+$

$\frac{2}{5} \Rightarrow 0$

12

[MCQ]

#Q. If 50! can be denoted maximum 7^x, then the value of x is?

A 9

B 8

C 7

D None of these

$\frac{50}{7} \Rightarrow 7$

$\frac{7}{7} \Rightarrow 1$

$\frac{1}{7} \Rightarrow 0$

$50! \Rightarrow 7^8$

[MCQ]

#Q. Find the number of zeroes in following multiplication:

Assignment

$5 \times 10 \times 15 \times 20 \times 25 \ldots\ldots \times 50$?

A 48

B 12

C 24

D 8

[MCQ]

#Q. How many squares are there in a chessboard?

Assignment



[MCQ]

#Q. What would be the unit digit in the answer of: $2362^{911} + 1418^{216}$

Assignment



2 mins Summary

Topic Number System

THANK - YOU